

Deepfakes, Voice Cloning & AI Social Engineering

The threat

AI makes two things real (or near-real) at commodity cost:

Voice clones – few minutes of audio → a convincing phone call from "your CEO"

Video deepfakes – low-quality but functional for Zoom-style social manipulation

Personalized phishing – AI crafts individualized messages from OSINT data; no more 'Dear Customer' mass blast

Case reality

Fraud calls using cloned executive voices asking for wire transfers (billions already lost in deepfake fraud – bank transfer scams using audio/visual clones)

Phishing emails with perfect local language + specific job context (AI does the context pull from LinkedIn)

Social engineering at scale: AI generates a convincing 'IT help desk' persona on a call

What still works: human process controls

Call-back verification (never act on an out-of-band request from voice alone)

Two-person rule for financial actions

Slow down (the adversarial psychology is speed + authority)

Simulate: run regular phishing/vishing exercises with the new AI-flavored lures

The defense is *process*, not tech

Tech defenses

Mark-origin protocols (email authentication, DMARC, voice origin marking – STIR/SHAKEN for caller ID)

Content provenance tracking (C2PA standard – provenance metadata)

Enterprise: verify who's calling via known-number confirmations; block spoofed caller ID

The honest view

AI social engineering is the most *practical* near-term threat: it scales, it fools humans reliably, and it costs almost nothing to create. The technical defenses lag; process + training is where orgs should invest now.

When AI writes the phish, the only reliable firewall is a trained, *slow*, process-following human who verifies out-of-band.

[← The AI Threat Landscape: Wha...](#)

[AI-Powered Malware & Evasion →](#)